

UNIT 1 • APPLICATIONS OF AI

Basic Concepts of Artificial Intelligence

Explore the definition, evolution, applications across industries, and the key benefits AI brings to our world.



Ch 1.1 — AI Overview



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What is Artificial Intelligence?



AI is the science of creating machines that can perform tasks requiring human-like intelligence — such as problem-solving, recognizing patterns, making decisions, and learning from experience.

How AI differs from traditional programs:

Traditional Programs

Operate on fixed, pre-written instructions. Cannot adapt or improve.

vs

AI Systems

Analyze data, adapt, and improve performance over time — even in new situations.

Expert Definitions

1

John McCarthy (Father of AI)

"Artificial Intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs." — Coined term "AI" at the Dartmouth Conference, 1956.

2

Yoshua Bengio (Deep Learning Pioneer)

"AI is the ability of a machine to learn from experience, adapt to new inputs, and perform tasks that normally require human intelligence." — Focuses on deep learning models that mimic neural networks.

A Brief History of AI



1940s–50s

Birth of AI

- Alan Turing proposed the Turing Test to evaluate machine intelligence (1950).
- John McCarthy coined "AI" at the Dartmouth Conference (1956).
- Marvin Minsky co-founded MIT's AI Lab.



1960s–70s

AI Winter

- High expectations were unmet due to limited computing power.
- Funding and interest in AI research declined sharply.



1980s–90s

Revival of AI

- Expert Systems were developed to mimic human decision-making.
- Commercial use grew in medical diagnosis and troubleshooting.



2000s–10s

Big Data Era

- Internet explosion + better hardware accelerated machine learning.
- AI used in speech recognition, image classification, recommendations.



2020s+

AI Everywhere

- AI transforming Healthcare, Agriculture, Education, Finance, Transport.

Key Milestones in AI

Year	Milestone	Description
1950	Turing Test	Alan Turing proposed evaluating machine intelligence by human-indistinguishable behavior.
1956	Term 'AI' Coined	John McCarthy introduced the term at the Dartmouth Conference.
1966	ELIZA	The first chatbot, simulating a psychotherapist using pattern matching.
1972	WABOT-1	First intelligent humanoid robot with vision and limb control, developed in Japan.
1980	Expert Systems	Programs like MYCIN helped doctors diagnose diseases.
1997	IBM Deep Blue	Defeated world chess champion Garry Kasparov.
2009	Waymo	Google launched its self-driving car project.
2011	IBM Watson	Won Jeopardy! by understanding natural language and vast databases.
2014	Pepper Robot	First robot to recognize human emotions.
2016	AlphaGo	Defeated Go champion Lee Sedol using deep reinforcement learning.
2018	Google Duplex	AI assistant that made phone calls on behalf of users, sounding human-like.

Types of Artificial Intelligence

AI is commonly classified into three levels based on capability:



Current Reality

Narrow AI

Also called Weak AI — designed for a specific task.

Examples: Siri, Alexa, Google Translate, Chess engines, Spam filters.

Most AI applications today fall into this category.



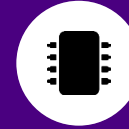
Near Future

General AI

Also called Strong AI — possesses intelligence comparable to human capabilities.

Can understand, learn, and apply knowledge across a wide range of tasks.

Does not yet exist in practice.



Theoretical

Superintelligent AI

Theoretical intelligence surpassing the best human minds in every field.

Could solve problems beyond human comprehension.

Raises major ethical questions.

Applications of AI in Daily Life

AI is woven into the everyday tools billions of people use.



Search Engines

AI algorithms (e.g. Google RankBrain) improve search relevance.



Voice Assistants

Siri, Alexa, Google Assistant understand and respond to voice commands.



Navigation Apps

Google Maps and Waze use AI to analyze traffic and suggest best routes.



Recommendation Systems

Netflix, YouTube, Amazon suggest content based on user behavior.



Health Apps

AI tracks fitness, detects heart conditions, and reminds medication schedules.



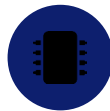
Biometric Security

Facial recognition and fingerprint scanning for unlocking phones and security.



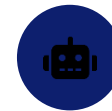
Language Translators

Google Translate converts speech or text into different languages instantly.



Social Media

AI filters spam, recommends content, and recognizes faces in photos.



Games & VR

AI opponents, virtual reality experiences, and real-time game decision-making.

Applications of AI in Industry



Healthcare

Medical Diagnosis

AI analyzes X-rays and MRIs to detect diseases like cancer and pneumonia.

Drug Discovery

Accelerates development of new drugs by predicting molecular behavior.

Personalized Treatment

ML models recommend treatments tailored to individual patient data.



Finance

Fraud Detection

Monitors transactions in real-time to identify suspicious activity.

Algorithmic Trading

AI systems analyze market data and execute trades at optimal times.

Risk Assessment

ML models evaluate the creditworthiness of borrowers.



Manufacturing

Predictive Maintenance

AI predicts equipment failures, reducing downtime and saving costs.

Quality Control

Vision systems inspect products faster and more accurately than humans.

Robotics

Industrial robots perform repetitive tasks like welding, assembling, packaging.

AI in Industry — More Sectors



Transportation

Self-Driving Vehicles

AI enables autonomous cars, trucks, and delivery drones.

Route Optimization

AI algorithms plan efficient delivery routes, saving time and fuel.

Predictive Shipping

ML predicts shipment delays and adjusts logistics planning.



Energy

Smart Grids

AI optimizes energy distribution and manages electricity loads intelligently.

Renewable Forecasting

AI predicts solar and wind energy production for better grid integration.

Equipment Monitoring

AI tracks the health of critical infrastructure like turbines and power plants.



Agriculture

Precision Farming

AI analyzes soil data, weather patterns, and crop health to maximize yield.

Autonomous Machinery

Self-driving tractors and drones monitor fields and spray pesticides accurately.

Livestock Monitoring

AI tracks animal health and optimizes feeding schedules.

Human Intelligence vs Artificial Intelligence

Feature	Human Intelligence	Artificial Intelligence
Source	Biological brain	Machine code and algorithms
Self-awareness	Fully self-aware and conscious	No self-awareness (yet)
Aim	Driven by emotions, survival, and values	Task-oriented, efficiency-driven
Functioning	Intuitive, emotional, creative	Logical, rule-based, fast
Learning	Learns from experiences, emotions, and logic	Learns from large datasets
Independence	Independent thought and decision-making	Needs human input for training
Adaptability	Adapts to unexpected situations creatively	Adapts only within training scope
Creativity	Can innovate and create original ideas	Can mimic creativity (art, music, text)
Ethics	Guided by human morals and conscience	Lacks ethics; depends on programming

Benefits of AI

AI delivers transformative advantages across every sector of society.



Automation

Automates repetitive tasks, saving time and reducing human effort across manufacturing, finance, and healthcare.



Accuracy & Precision

Performs tasks with high accuracy (medical diagnosis, quality control) and reduces human errors significantly.



24/7 Availability

AI systems never sleep — ensuring continuous operation for customer service, chatbots, and monitoring systems.



Data Processing & Analysis

Processes vast amounts of data quickly, uncovering insights impossible for humans to find manually.



Decision Making

Assists in complex decisions — e.g., doctors in surgery, financial advisors with robo-advisor tools.



Remote Patient Monitoring

AI wearables and health apps track vital signs, detect conditions early, and alert doctors remotely.



Solving Complex Problems

AI tackles multi-variable problems — climate models, drug discovery, logistics optimization at scale.



Enhancing Human Abilities

Amplifies human potential — AI tools help pilots, surgeons, engineers, and researchers perform better.

Disadvantages of AI

Despite its power, AI comes with significant challenges that need to be addressed.



High Initial Cost

Developing and implementing AI systems is expensive — technology, infrastructure, and expertise.



Job Displacement

AI automation can replace human workers, especially in routine, low-skill jobs.



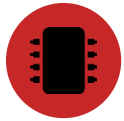
Lack of Creativity

AI can solve data-based problems but lacks human creativity, emotions, and critical thinking.



Data Privacy & Security

AI systems rely heavily on data, raising concerns about personal information collection and storage.



Dependence on Technology

Over-reliance on AI could make humans less skilled or less critical in certain fields.



Bias & Ethical Concerns

AI can inherit biases from training data, leading to unfair or unethical decisions.

Chapter Recap



Definition:

AI is the science of creating machines that can think, learn, and adapt — going beyond fixed-instruction programs.



Evolution:

From Turing's 1950 test → AI Winters → Expert Systems → Big Data → 2020s: AI is now everywhere.



Types of AI:

Narrow AI (current): task-specific | General AI (future): human-level | Superintelligent AI (theoretical): beyond humans.



Daily Life Apps:

Voice assistants, recommendation engines, navigation, health apps, biometric security, language translation.



Industry Apps:

Healthcare (diagnosis), Finance (fraud detection), Manufacturing (robotics), Transport (self-driving), Agriculture (precision farming).



Benefits:

Automation, precision, 24/7 availability, data analysis, remote patient monitoring, decision support, solving complex problems.



Disadvantages:

High cost, job displacement, lack of creativity, data privacy risks, technology dependence, bias & ethical concerns.